

Richard Alexander Pedraza Martínez

Senior Backend Engineer | Python · Django · AWS · AI & Automation

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 8+ years of experience designing and scaling robust backend systems with Python, Django, FastAPI, and Node.js. Specialized in AWS-based architectures, advanced database management (PostgreSQL, Redis), and CI/CD pipelines, with a focus on high availability and performance. In recent years, I have led the development of production AI solutions—from real-time voice agents (Pipecat, LLMs) to complex process automation—cutting processing times by up to 70% and serving critical multi-tenant workloads. Experienced in leading teams and defining engineering standards, translating business needs into high-impact technical solutions.

PROFESSIONAL EXPERIENCE

Senior Backend Engineer — Disglobal

Remote · Aug 2025 – Present

- Lead the design and development of the entire backend for an insurtech mobile platform that aggregates multiple insurance providers, enabling users to quote and purchase policies through a single unified experience.
- Designed a **provider-agnostic architecture** that abstracts each insurer's API behind a common interface; currently 5 insurers integrated in production, each with its own API.
- Built the backend with **Django REST Framework, PostgreSQL, and Redis**, orchestrating quoting, underwriting, and policy issuance flows, and optimizing performance through caching.
- Implemented **automated document OCR with AWS Textract**, streamlining the capture and validation of policy data and user documentation.
- Built a **multi-vendor pharmacy marketplace** module, managing catalogs, inventory, and orders for multiple pharmacies within the app.
- Integrated AI-powered health tools from external providers, extending the platform's capabilities beyond insurance.

Lead AI Voice Engineer — Tesla Media Group

Remote (California) · Aug 2025 – Present

- Built an end-to-end **production AI voice phone assistant** serving **10 distinct clients (multi-tenant)** and handling **5,000+ call minutes per month**, on a **4-microservice Docker architecture** (FastAPI, Express, NestJS, and a headless browser).
- Orchestrated a **real-time voice pipeline with Pipecat** (Deepgram STT → OpenAI GPT-4.1-mini LLM → Cartesia TTS) achieving **sub-800ms response latency**, with barge-in/interruption handling, end-of-turn detection, and VAD, bridging telephony across PSTN ↔ WebRTC ↔ WebSocket.
- Designed an **API-driven dynamic conversational flow engine** that replaced hard-coded flows, generating dialog nodes and LLM function handlers at runtime from JSON definitions, with episodic memory, multi-tenant identity, and prompt-injection protection.
- Developed a **telephony broker** that orchestrates a headless Chromium instance (Playwright) running RingCentral's SIP/WebRTC WebPhone, with audio relay and a custom PCM/WAV frame serializer.
- Implemented real-time observability (live transcription, TTFB/token metrics) and automated production deployment via SSH/rsync with per-environment credential injection.

Remax

Remote · Sep 2023 – Jan 2026

Tech Lead

Jan 2025 – Jan 2026

- **Led a team of 2 developers**, owning the feature roadmap, prioritizing tasks, and tracking delivery progress.
- Led the architecture and management of a business-critical multi-service platform on **AWS** (compute, databases, serverless, automation, and analytics), ensuring high availability and operational continuity.
- Established engineering standards (security, observability, code quality, and deployments) and technical governance that reduced technical debt and accelerated delivery speed.
- Implemented **CI/CD pipelines with Jenkins**, standardizing deployments and reducing production errors and downtime.
- Managed **PostgreSQL** in production (performance tuning, migrations, security controls) and led backend development in **Python and Node.js**.

Full-stack Developer

Sep 2023 – Jan 2025

- Delivered an **automated property valuation system** processing **5,000+ listings per cycle**, cutting valuation time by **70%** and reaching up to **90% pricing accuracy**.
- Built full-stack web applications with REST APIs (**Django, FastAPI**) and a frontend in **Next.js, React, and Tailwind**.

- Developed large-scale data extraction pipelines (**Scrapy**, **BeautifulSoup**) and processing pipelines with **Pandas** and comparative pricing models.
- Integrated **n8n** automations for lead management, operations, and reporting, eliminating significant manual workloads.
- Built interactive dashboards providing real-time visibility for internal teams.

Senior Consultant — Backend & AI (Freelance)

Remote · Feb 2025 – Dec 2025

Audio transcription platform (client under NDA · Madrid)

- Architected the backend for a high-volume automated audio transcription SaaS platform (a product similar to Plaud, with no hardware involved).
- Orchestrated transcription jobs using **asynchronous task queues**, optimizing throughput for large audio workloads.
- Designed a privacy-first authentication mechanism with **irreversible (zero-knowledge) hashing** for secure storage of sensitive identifiers.

Voice-based recruitment intelligence platform (client under NDA · Miami)

- Architected AI agents for a system that **ranks talent and delivers voice summaries to recruiters**, combining conversational flows with large-scale data analysis.
- Implemented a **classification engine processing 6,000+ metadata-rich records** under dynamic evaluation rules.
- Built **asynchronous batch-processing** architectures for continuous parallel workloads and established system observability standards.

Automation Engineer — Crazy Imagine Software

Remote · Mar 2025 – Jun 2025

- Designed and deployed complex **n8n** automations for business-critical B2B workflows, integrating chatbots, **LLM** pipelines, and various SaaS APIs, enhanced with **Python and JavaScript** scripts to build event-driven flows.
- Synced data across **CRM, ERP, and marketing platforms** through internal and third-party API integrations, improving data accuracy and operational efficiency.
- Implemented AI capabilities (LLM APIs, **vector search**, and prompt engineering) for content generation and analysis within automation pipelines.

Integration & Automation Specialist — Universidad ISEP

Remote · Jan 2025 – Mar 2025

- Designed and optimized automated workflows using **no-code/low-code tools (n8n)** and API integrations, connecting disparate systems and reducing manual tasks.
- Analyzed business requirements to develop custom automation solutions, driving digital transformation across multiple operational units.

TECHNICAL SKILLS

Languages: Python · Node.js · TypeScript · JavaScript · SQL · Bash

Backend & APIs: Django · Django REST Framework · FastAPI · Express.js · NestJS · REST APIs · async · async task queues

AI & Voice: Pipecat · LLMs (OpenAI / GPT-4.1) · Deepgram (STT) · Cartesia (TTS) · conversational agents · vector search · prompt engineering · real-time voice pipelines

Databases: PostgreSQL (AWS RDS) · Redis · SQLite · MySQL (familiar)

Cloud & DevOps: AWS (EC2, Lambda/serverless, RDS, S3, Textract) · Docker · Docker Compose · CI/CD · Jenkins · SSH/rsync

Real-Time & Telephony: WebSockets · WebRTC · SIP · RingCentral SDK · Playwright

Data & Automation: n8n · Pandas · Scrapy · BeautifulSoup · web scraping · ETL pipelines · API integration (CRM/ERP)

Frontend: React · Next.js · Tailwind CSS

EDUCATION

Computer Engineering — Instituto Universitario de Tecnología (I.U.T.A.I)

2010 – 2016